

Real-time structural damage localisation using modal curvatures based on Machine Learning algorithm

23 September 2026

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□ Structural Health Monitoring (SHM)

✓ Non-destructive test methods (local)

- Magnetic Particle Testing
- Laser Testing
- Digital Image Correlation
- Radiography Testing
- Ultrasonic Testing



✓ Vibration-based methods (global):

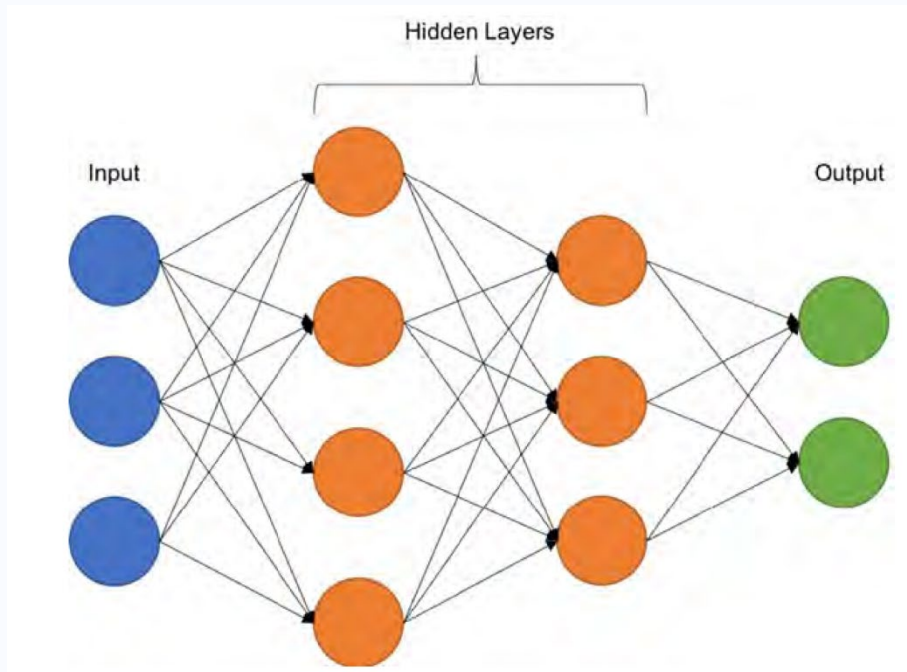
- Natural frequency-based method
 - Modal curvature-based method
- } Reliable approach



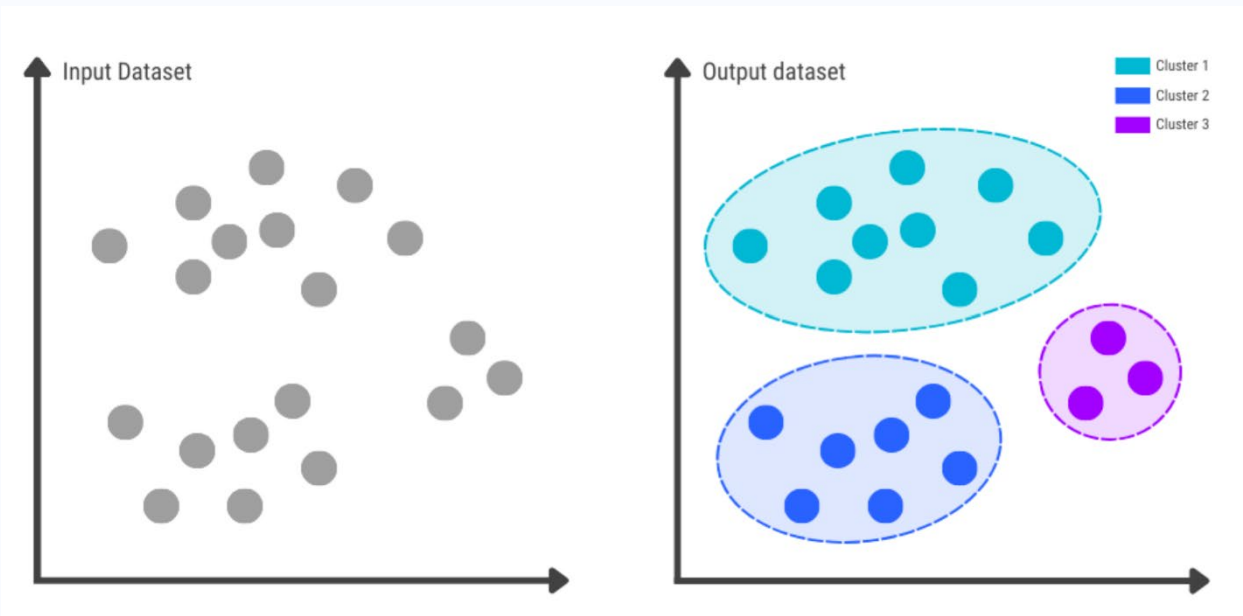
❑ Machine Learning (ML):

- A multi-sensor system is ideal for SHM,
- However, such systems generate vast data volumes,

✓ ML algorithm

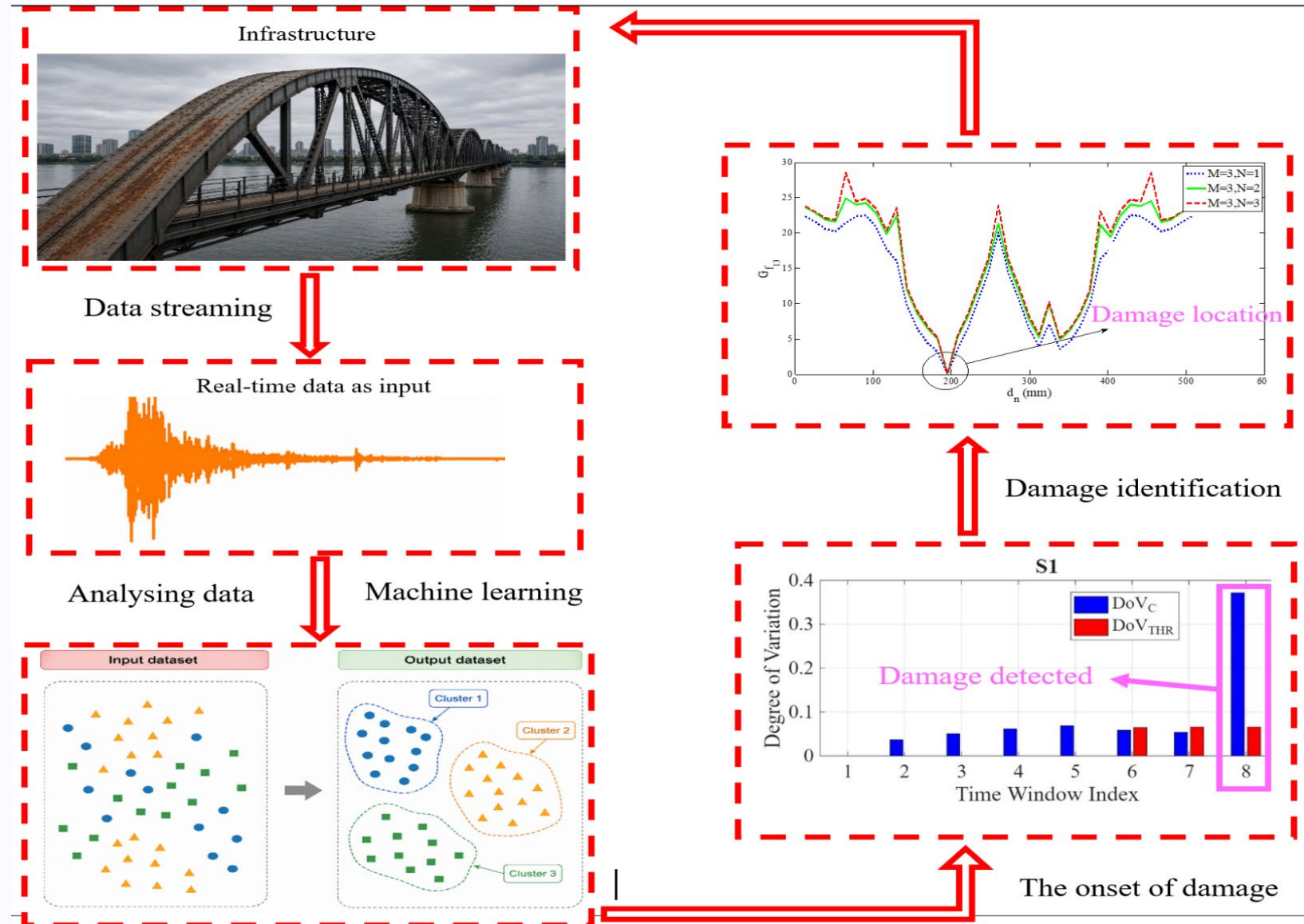


✓ Unsupervised ML

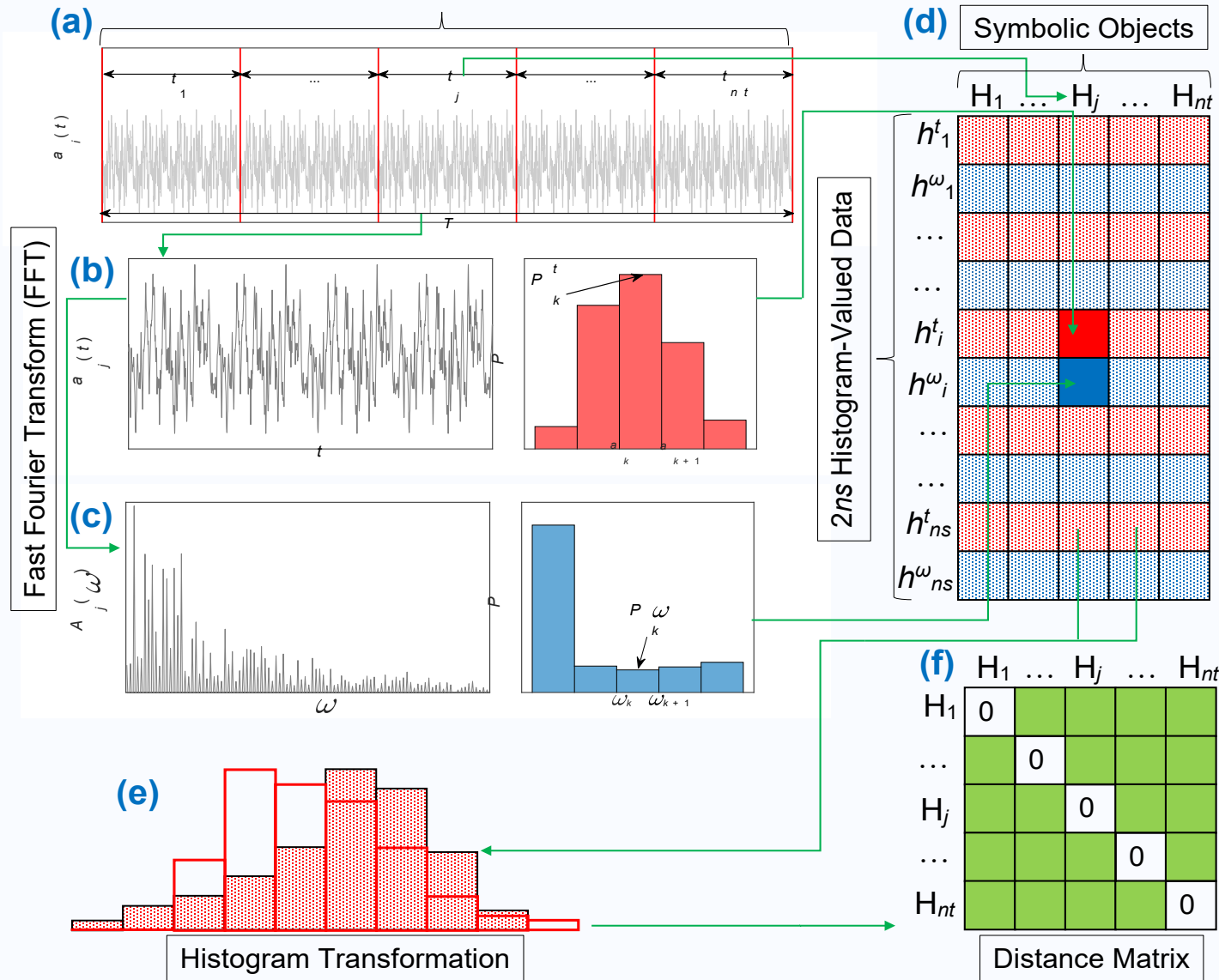


The present study

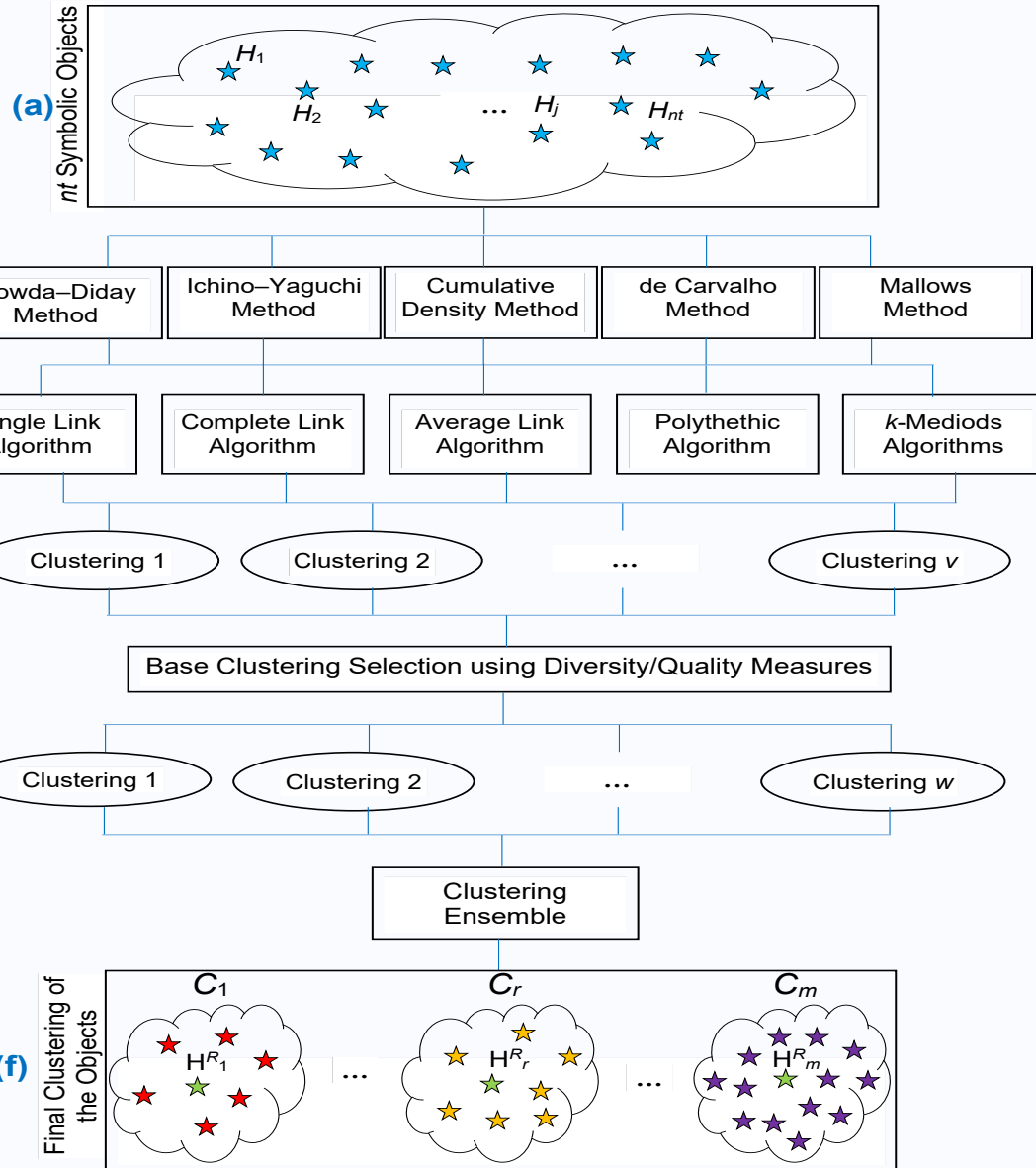
This work introduces a real-time structural damage localisation based on ML algorithm using modal curvatures which is novel in ML-SHM applications



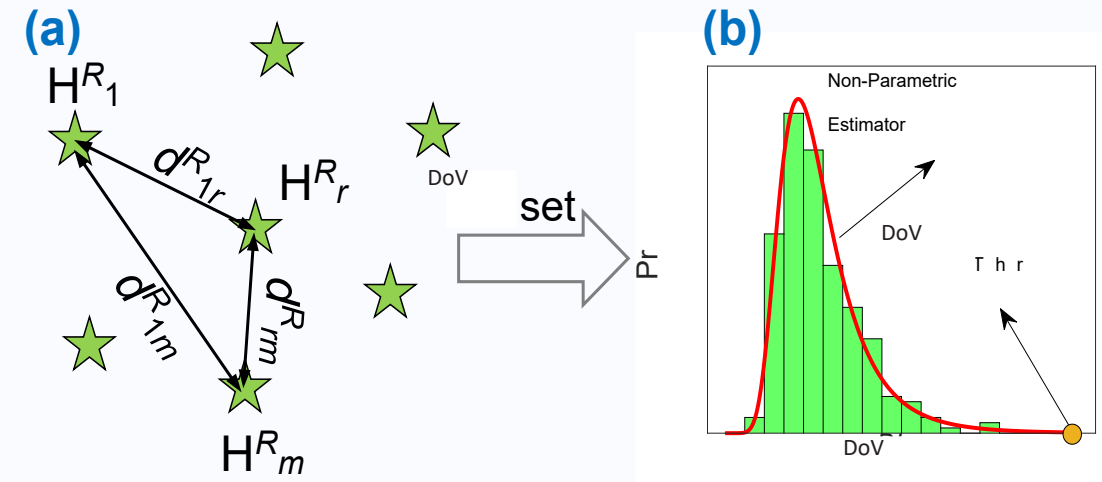
Dimensionality Reduction

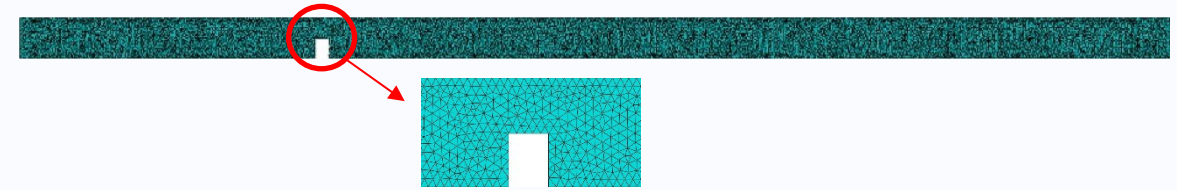
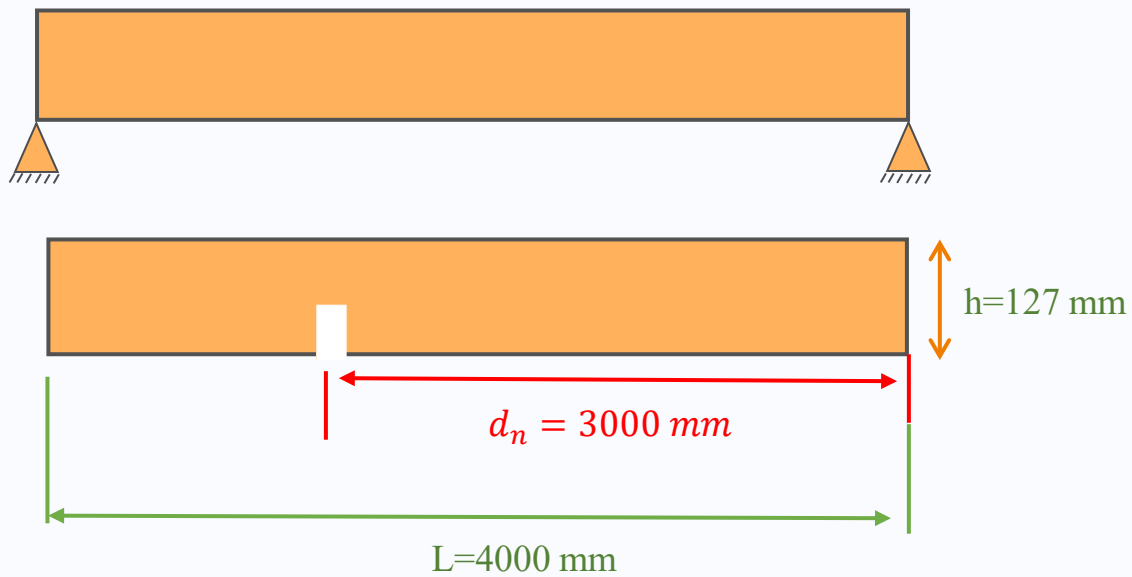


Variation Quantification



Anomaly Detection

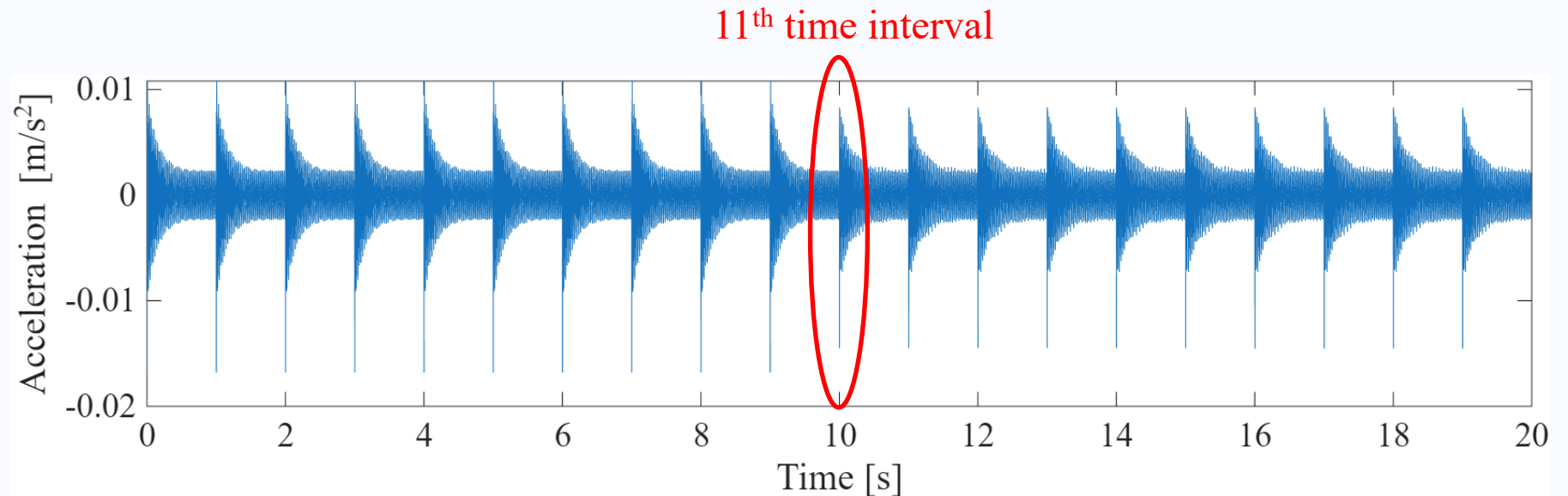




The FEM of beam (mesh size 0.01m)

Schematic of damaged beam longitudinal view

Time-acceleration data



Equation of motions

$$\frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = \rho \ddot{u}_x$$

$$\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_y}{\partial y} = \rho \ddot{u}_y$$

Hooke's law

$$\sigma_{ij} = C \varepsilon_{ij}$$

Harmonic solution

$$u_i(x, y, t) = U_i(x, y) \cos(\omega t - \alpha)$$

Governing equations

$$\left\{ \begin{array}{l} \mu \nabla^2 U_x(x, y) + \frac{E}{2(1-\nu)} \frac{\partial}{\partial x} \left(\frac{\partial}{\partial x} U_x(x, y) + \frac{\partial}{\partial y} U_y(x, y) \right) = -\rho \omega^2 U_x(x, y) \\ \mu \nabla^2 U_y(x, y) + \frac{E}{2(1-\nu)} \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} U_x(x, y) + \frac{\partial}{\partial y} U_y(x, y) \right) = -\rho \omega^2 U_y(x, y) \end{array} \right.$$

Simply supported boundary conditions

$$U_x = U_y = 0 \quad \left\{ \begin{array}{l} X = 0 \\ X = L \end{array} \right.$$

Modal curvatures

$$\chi = 2\varepsilon_{xx}/h$$

Number of Time-Domain Bins

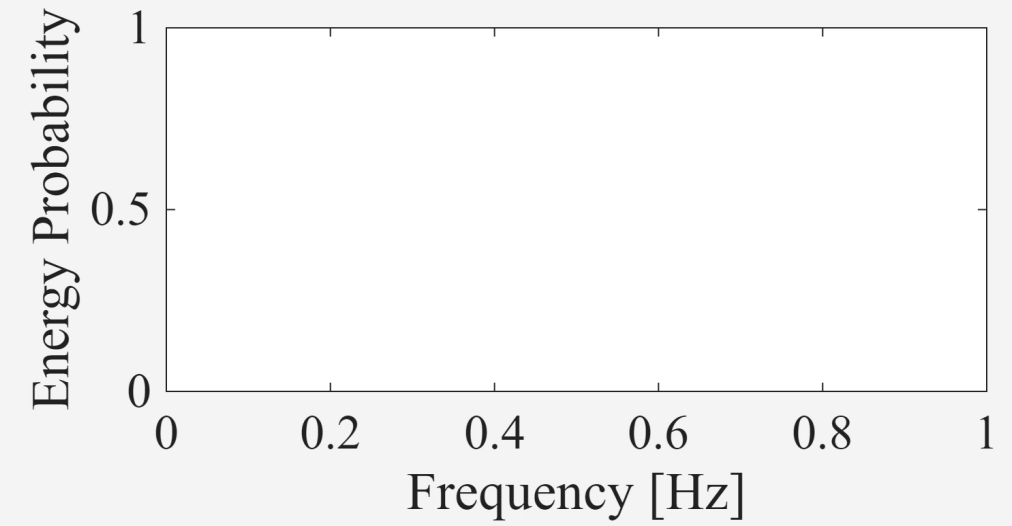
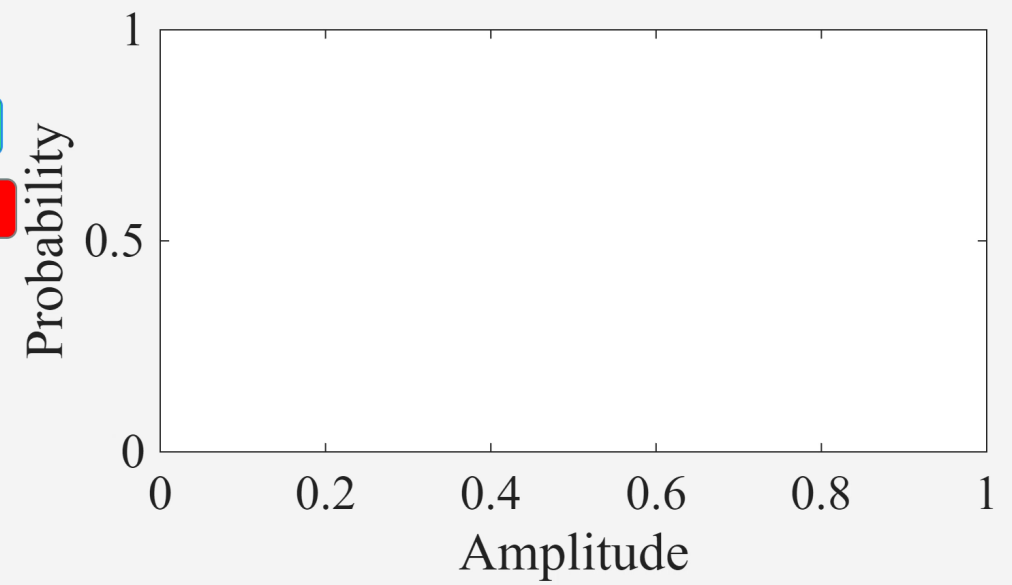
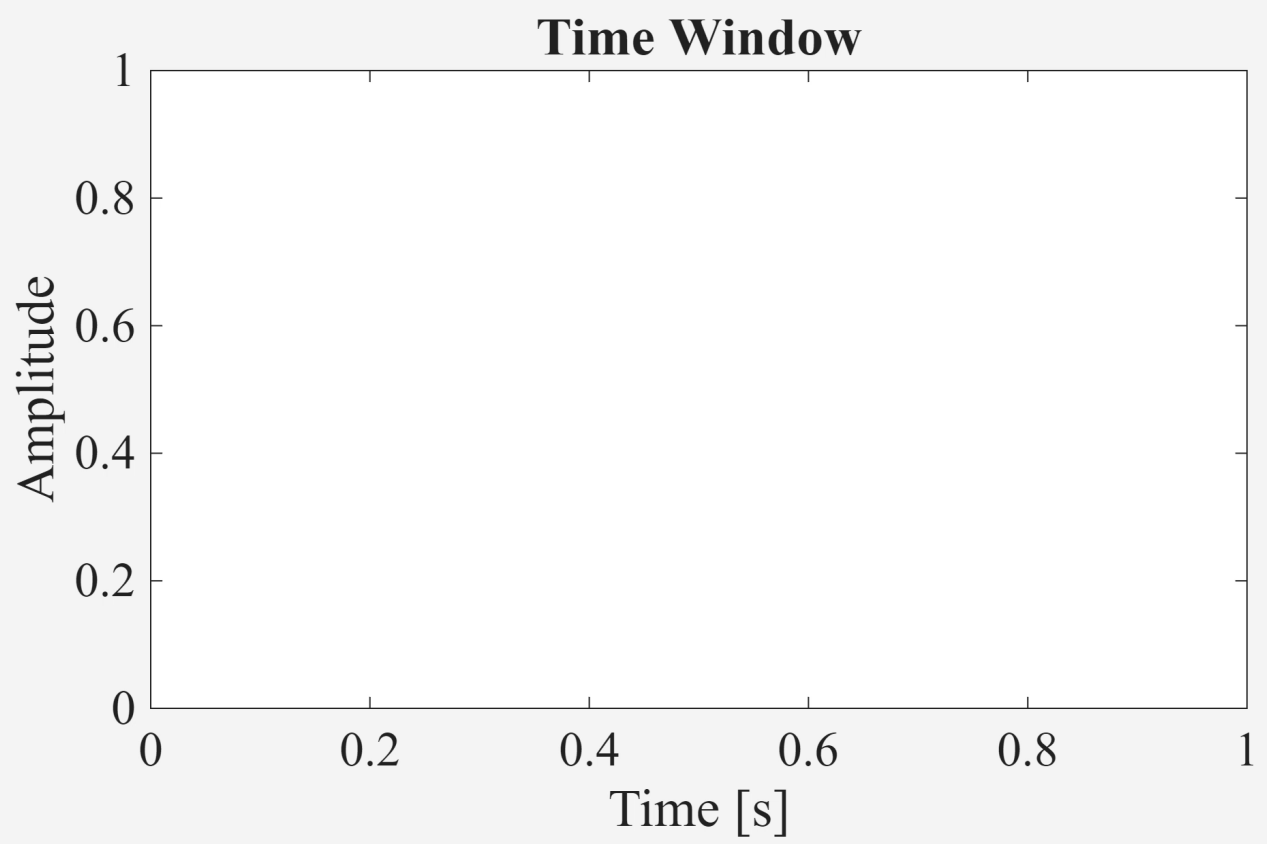
Number of Frequency-Domain Bins

Number of Time Intervals Per Window

Time Interval Length [s]

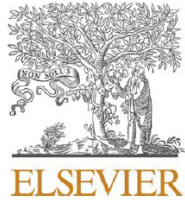
Cutoff Frequency (Hz)

Plot Sensor Data





Experimental tests



Contents lists available at ScienceDirect

Reliability Engineering and System Safety

journal homepage: www.elsevier.com/locate/ress**IF=13.7****H-Index=209**

A self-adaptive mathematical approach for real-time structural health monitoring and anomaly detection using unsupervised machine learning

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ARTICLE INFO

Keywords:

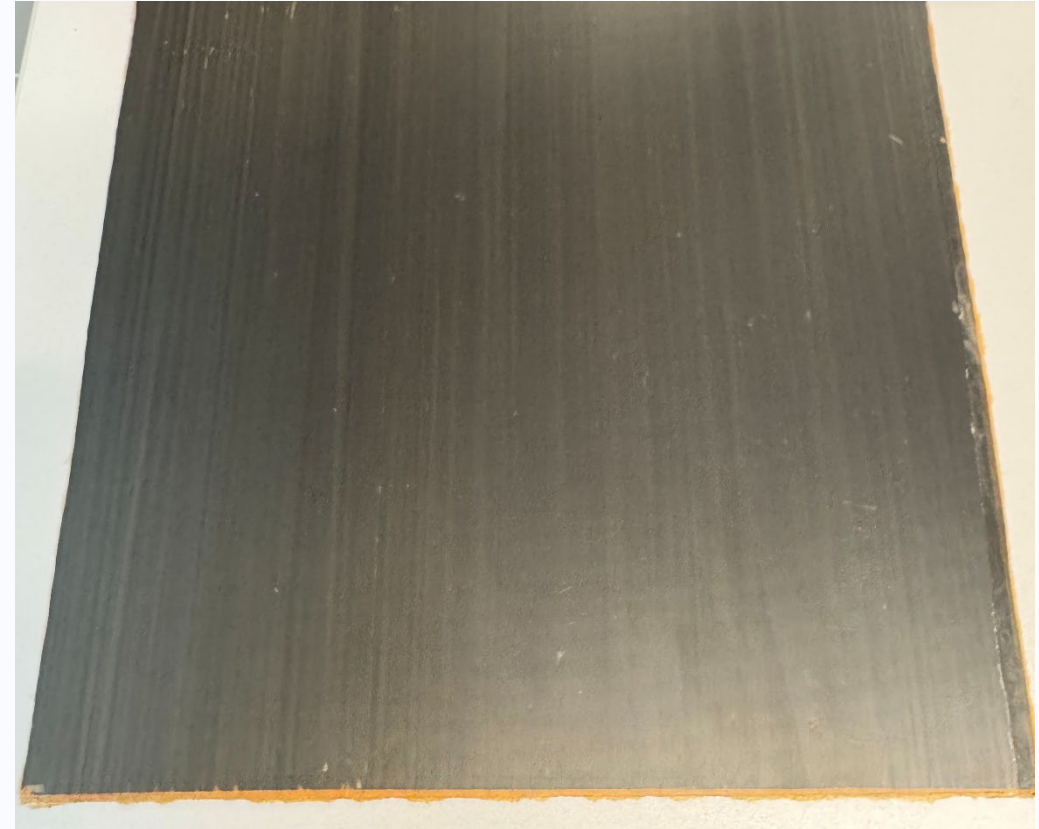
Real-time data processing
Unsupervised machine learning
Structural health monitoring
Damage detection
Asset management

ABSTRACT

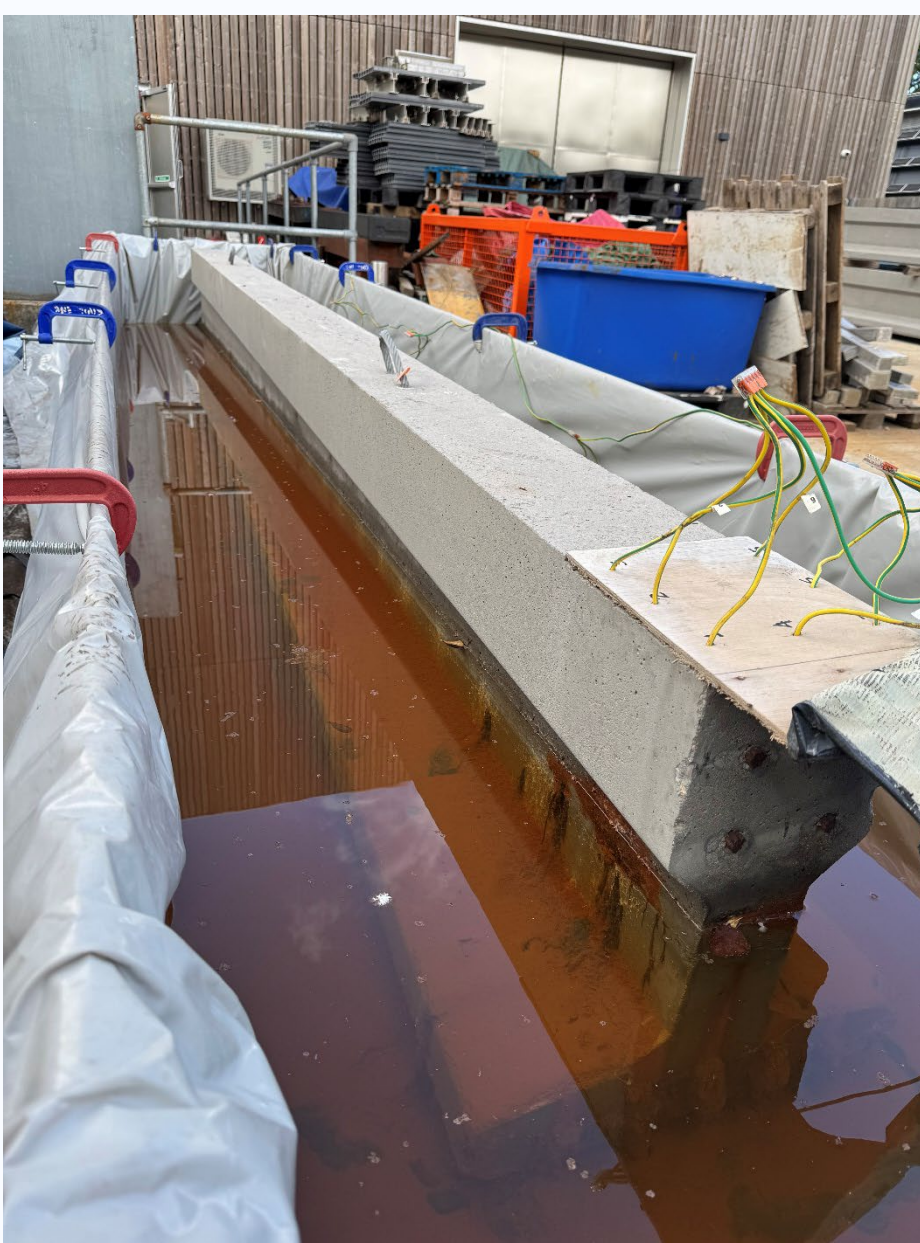
With recent advancements in state-of-the-art sensing technologies, dense hybrid sensor networks enable integration of numerous multi-type sensing tools, empowering real-time monitoring of assets. Such sensor networks have significantly increased damage sensing capability and reliability at global and local levels, empowering real-time monitoring for intelligent asset maintenance. Such real-time monitoring, however, produces large data that exceeds the present knowledge and capacity to rapidly analyse and accurately discover their feature for structural damage detection, a challenge often referred to as “data deluge”. To meet this challenge, this study establishes a new precise, generalised, automatic mathematical framework that intelligently tracks and analyses variations in large data produced from dense sensor networks in real-time. Such framework systematically integrates modern data techniques for anomaly or damage detection. This study proposes a novel mathematical framework for unsupervised machine learning, real-time structural health monitoring that enables adaptive damage detection directly from streaming vibration data without requiring labelled damage data or physics-based numerical models. By integrating symbolic data analysis, unsupervised learning, and adaptive thresholding into a unified framework, the proposed approach provides a general solution for automated anomaly detection and warning generation. In the proposed framework, exceeding the adaptively learned threshold prompts an instantaneous damage alarm. The proposed framework is validated using both numerical beam models and experimental tests on corroded and uncorroded columns, demonstrating its capability for accurate and fully automated real-time damage detection under both simulated and experimental conditions.



Small concrete beams



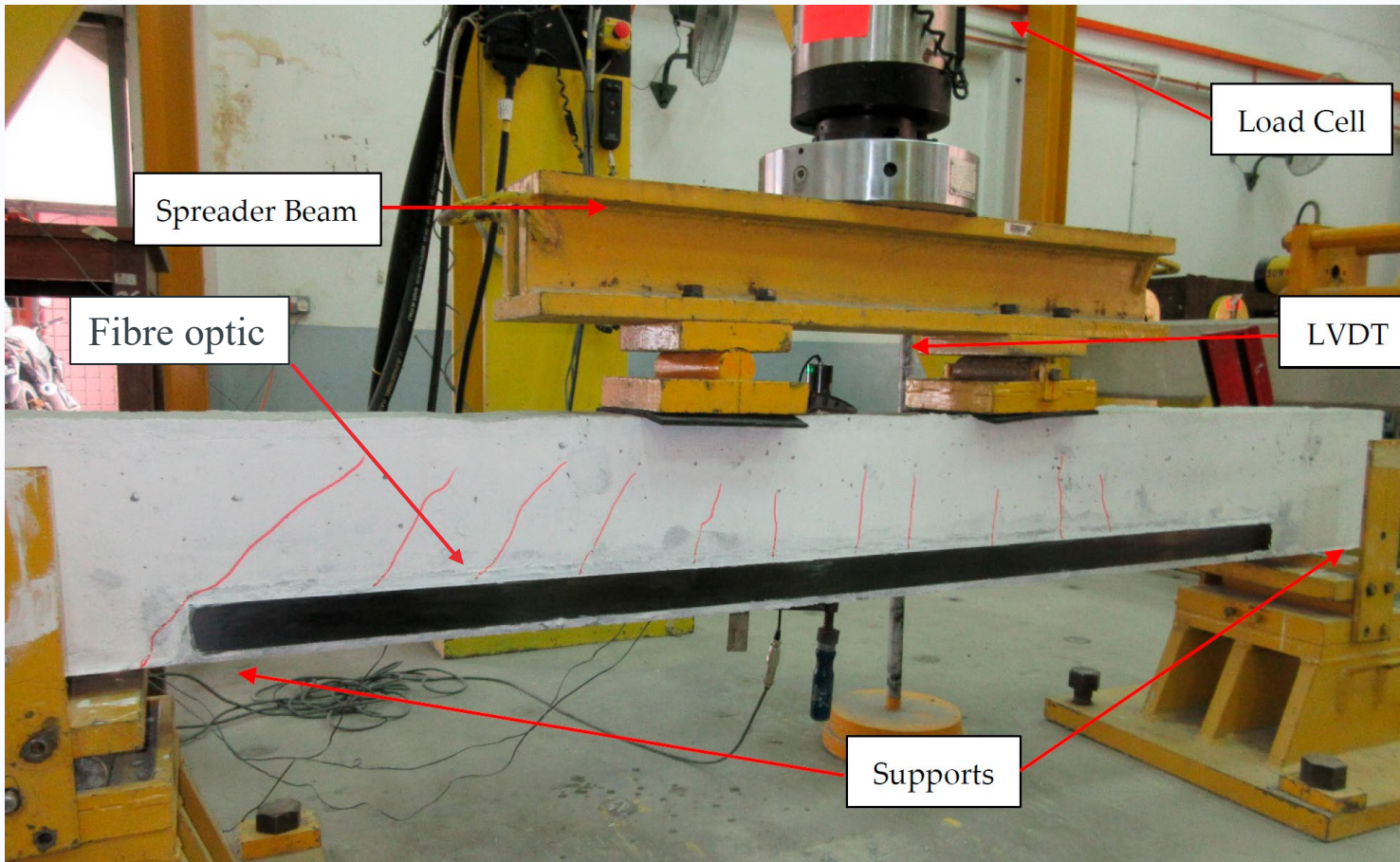
Composites (CFRP/GFRP)



Corroded prestressed concrete beam



Uncorroded prestressed concrete beam



Four-point bending test



Impact test

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